

Zhiyao He

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Robot Learning · Robot Manipulation · Reinforcement Learning · Robotics Simulation · Computer Vision · LLM Agents

Profile

I am an AI engineer working on embodied AI systems. I build robot pipelines that connect perception, language models, planning and control, from simulation (Isaac Gym, MuJoCo, Gazebo) to real hardware. My recent work is a voice-controlled navigation and manipulation system on the Unitree Go2-W quadruped robot. I am seeking a PhD position in robot learning, especially on how robots can learn manipulation skills that work reliably in the real world.

Education

M.S. in Signal and Information Processing Hohai University, Nanjing, China Sep 2015 – Jul 2018

Grade: 82.33 / 100 (research-based degree)

- Thesis work on machine learning for runoff forecasting; published two peer-reviewed journal papers and obtained one Chinese invention patent.
- Relevant courses: Information Acquisition and Processing, Optimization. Certificate: Deep Learning Specialization (DeepLearning.AI).

B.Eng. in Electrical Engineering and Automation Anhui Agricultural University, Hefei, China Sep 2011 – Jul 2014

CGPA: 3.65 / 4.00

- Finished the four-year program in three years.
- Relevant courses: Control Theory, Circuit Design, C Programming, Digital Signal Processing.

A.S. in Computer Science University of the People, USA Nov 2021 – Nov 2024

CGPA: 4.00 / 4.00

- President's List for 12 terms.
- Relevant courses: Python, Java, Databases, Operating Systems, Software Engineering, Computer Networks.

Publications and Patent

- Zhiyao He, Mingxiang Yang, Hao Wang, et al. "Annual Average Runoff Ensemble Forecast for Jinping I-Stage Hydropower Station Based on Elman Neural Network." Water Resources and Power, 2017.
- Chenming Li, Lei Zhu, Zhiyao He, et al. "Runoff Prediction Method Based on Adaptive Elman Neural Network." Water, 2019.
- Zhiyao He (Patent holder). "Medium- and Long-term Runoff Forecasting Method Based on Mutual Information, Kernel PCA, and Elman Neural Network." Chinese National Invention Patent, CN107463993B, granted 2020.

Work Experience

AI Engineer WellWin Technology Co., Ltd., Hengqin Island, China Mar 2026 – Present

- Built the perception, planning and control stack for voice- and text-controlled navigation and manipulation on the Unitree Go2-W quadruped robot with a D1 arm (details in Projects).
- Implemented the closed-loop control system of the AKI humanoid robot: a reinforcement learning policy layer, a joint target layer, and PD motor control.
- Designed the AKI robot's power management system, covering power input, distribution and conversion, and the load execution layer.
- Integrated YOLO-based perception with ASR, LLM and TTS modules for face recognition and voice interaction.
- Set up simulation environments and ran locomotion training for the AKI and Unitree G1/R1 humanoid robots in Gazebo, MuJoCo, and Isaac Gym.

Independent Study and Research Nanjing, China Jul 2019 – Feb 2026

- After Huawei, I planned to return to research. During this period my second journal paper (Water, 2019) was published and my invention patent was granted (2020).
- From 2021 to 2024, I completed a full computer science degree at University of the People (GPA 4.00/4.00, see Education) to retrain from signal processing toward AI and robotics. Part of this period overlapped with the COVID-19 lockdowns in China.

- Admitted to the M.Sc. in Data Science at Trinity College Dublin (2025 intake), but did not enroll.
- After that, I prepared for AI engineering interviews and joined WellWin Technology in March 2026.

Software Engineer Huawei Technologies Co., Ltd., Nanjing R&D Institute, China Jul 2018 – Jun 2019

- Worked on computer vision systems for face and vehicle recognition, mainly on development and performance optimization.
- Debugged and tuned the systems together with other teams.

Research Intern China Institute of Water Resources and Hydropower Research, Beijing, China Jun 2016 – Jun 2017

- Built a complete runoff prediction pipeline: data collection and cleaning, feature engineering, model development, hyperparameter tuning, training and evaluation.
- This work produced two journal papers and one invention patent (listed above).

Selected Projects

Voice-Controlled Navigation and Manipulation on Unitree Go2-W

- Perception: YOLO11 object detection with a wrist-mounted depth camera, plus 2D-to-3D coordinate transformation to locate the target in the world frame.
- Navigation: SLAM-based autonomous navigation to the target position.
- Language: a task planning pipeline with ASR, LLM, OpenClaw and TTS, so the robot can be controlled by natural language and reply by voice.
- Manipulation: a grasping pipeline based on forward/inverse kinematics and Jacobian-based motion control.
- Developed in ROS 2, tested in MuJoCo simulation and on the real robot.

AKI Humanoid Robot

- Real-time control loop for locomotion and interaction: an RL policy generates joint positions, a joint target layer does scaling, integration and clipping, and a PD layer drives the motors.
- ASR–LLM–TTS pipeline for natural voice interaction.

Voice RAG Assistant on AgiBot Lingxi X2

- An in-store retail assistant: customers ask about products, plans, and pricing by voice.
- Built a fully offline pipeline on a Jetson for events with unreliable networks: a bridge node streams the X2's microphone audio to a local FastAPI backend, which runs local ASR (FunASR/Paraformer-zh), a local LLM (Qwen2.5-7B via Ollama), pgvector RAG over the product database (PostgreSQL), and local TTS (Sherpa-ONNX / Piper).

Technical Skills

- Programming & Systems: Python, C++, Linux, Docker, Git
- Robotics: ROS/ROS 2, Isaac Gym, Gazebo, MuJoCo, SLAM, Navigation, Localization, Perception
- AI & Robot Learning: PyTorch, Machine Learning, Computer Vision, Reinforcement Learning, ASR, TTS, LLMs
- Robot Platforms: Unitree G1, Unitree Go2/Go2-W, AgiBot Lingxi X2

Research Interests

- Robot Learning and Reinforcement Learning
- Vision-Language-Action Models
- Robot Manipulation and Grasping
- Robotics Simulation, SLAM and Navigation

Honors and Awards

- President's List, University of the People (12 terms) | 2021–2024
- Outstanding Contribution Award, Huawei Nanjing R&D Institute | 2019
- Outstanding Graduate Student, Hohai University | 2017
- Hohai University Scholarship | 2015–2017
- Anhui Agricultural University Scholarship | 2012

Teaching and Outreach

Humanoid Robotics Curriculum on AgiBot Lingxi X2 WellWin Technology

- Designed a humanoid robotics curriculum for Macao secondary school students, progressing from no-code tools (LingXin, LinkCraft) to ROS 2 programming with the AimDK SDK, and finally to multi-platform integration.

Humanoid Robot Task Challenge, MYAIR 2027 Macau Youth AI and Robotics Competition

- Designed the task and rules for a humanoid robot ball-shooting competition, testing vision recognition, trajectory planning and 3D printing.

UoPeople Ambassador University of the People (AY2022, Term 5)

- Answered prospective students' questions about admissions and academics, helping applicants with limited educational opportunities gain access to quality higher education.